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United Nations Development Programme

PROJECT DOCUMENT

[Lebanon]

Project Title: Sustainable Energy for Public Services entities, entrepreneurship and innovation– EU Support to Lebanon's Green Agenda (CEDRO 6)
Project Number: 01001256
Implementing Partner: UNDP
Start Date: 01 November 2023 **End Date:** 30 October 2027 **PAC Meeting date:**

Brief Description
This project will provide solar energy solutions to the public sector, thus ensuring the continued provision of essential services and to bridge the energy gap that the national utility is unable to meet now. By doing so, the cost of electricity on the national budget is also lowered which would alleviate the financial burden on both the central government and on the sectors themselves, allowing them to use the financing for other essential services that are needed. On a global level, the Action is also aligned with the Lebanese unconditional National Determined Contribution (NDC) under the Paris Agreement to reduce by 20 % the Greenhouse gas emissions (GHG) by 2030, increase the renewable energy to reach 18% of the electricity produced by 2030 and a 3 % energy efficiency target by 2030 The project strategy consists of approaching the transition to renewable energy in Lebanon at two levels: one is to provide and improve access to energy for the public sector, including the Lebanese Armed Forces and the Internal Security Forces and the second is to support the private sector to upscale solar energy supply to greening recovery and creating jobs. Parallel to these two approaches, improving capacities, policies and legislation to push the sector forward will also be tackled based on the lessons learnt and needs identified during implementation at field level of both components. Ultimately, the overall impact of the project is the improved energy security and transition to sustainable energy and resource efficiency in Lebanon. This will be reached based on the below theory of change and in the later sections, more details are provided about the detailed outcomes, outputs and activities.

Contributing Outcome (UNSDCF,CPD): UNSDCF: 1.2.b (SDG output 13.2) Reduction of CO2eq emissions, through UN support (tonnes) CPD Outcome 3: Strengthened stabilization and green recovery to reduce vulnerabilities and environmental risks, including through enhanced competitiveness and business environment of sustainability-oriented MSMEs and high potential green productive sector values chains. CPD output 3.2 (IRRF output 5.2): Transition to renewable energy accelerated capitalizing on technological gains, clean energy innovations and new financing mechanisms to support green recovery. CPD output indicator: 3.2.1. Amount of renewable energy installed annually (output indicator for UNSDCF output 1.2, IRRF 5.2.1. Increase (in megawatt) in installed renewable energy capacity with solar, wind, biomass and other technologies) Indicative Output(s) with gender marker: Output 2.2.2 Number of women engineers empowered with EU support (GEN2)

Total resources required:	USD 24,321,900.00	
Total resources allocated:		
	UNDP 888:	USD 99,900
	Donor (EU):	EUR 22,000,000 equivalent to USD 24,222,000.00
	Government:	-
	In-Kind:	-
Unfunded:	-	

Agreed by (signatures):

UNDP
Ms. Melanie Hauenstein **Resident Representative** 22-Sep-2023 DocuSigned by: Melanie Hauenstein 79642E8B91B7414...
Date:

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I. DEVELOPMENT CHALLENGE

Background

Lebanon is enduring a severe and prolonged economic depression. According to the Fall 2021 World Bank (WB) Lebanon Economic Monitor (LEM)[1], the economic and financial crisis is potentially ranking as top 3 most severe economic collapse globally since the mid-nineteenth century. In the face of such colossal challenges, continuous policy inaction and the absence of a fully functioning executive authority to implement reforms threaten already dire socio-economic conditions and a fragile social peace.

For over two years, Lebanon has been facing compounded challenges: its largest peace-time economic and financial crisis, COVID-19 and the Port of Beirut explosion. However, Lebanon's leadership policy responses to these challenges have been highly inadequate. This is not so much due to knowledge gaps or lack of quality advice, but rather the result of a dysfunctional governance system based on vested interests, which hampers the achievement of political consensus over effective policy initiatives. With a history of a prolonged civil war and multiple conflicts— Lebanon is identified by the World Bank as a Fragility, Conflict and Violence (FCV) State - there is growing wariness of potential triggers to social unrest.

The poor quality of public services and inefficiencies in the public administration result in loss of time and resources by citizens and businesses alike and provide ample opportunities for corruption. The country's political system is designed to reinforce multi-confessional co-existence and gives way to clientelism and deeply rooted vested interests. Corruption is widespread in public and private sectors and infiltrates all levels of society, as reflected by the country's global and regional average performances scores in most areas of governance. In addition, the limited role played by civil society at policy level renders the adoption and implementation of key government reform policies difficult. The May 2022 parliamentary elections also resulted in a more fragmented Parliament, with the absence of a clear majority and growing support for alternative candidates representing civil society at the expense of traditional parties.

Real GDP has decreased by 10.5% in 2021, following a 21.4 % contraction in 2020. This represents the highest contraction in a list of 193 countries. The national currency, the Lebanese Pound (LBP), lost around 90% of its value against the US dollar since the end of 2019, which has caused triple-digit inflation as most production inputs and consumption goods are imported from abroad.

As a result, over half of the population is likely below the national poverty line, with the bulk of the labour force - paid in Lebanese Pound (LBP) - suffering from plummeting purchasing power. Increased numbers of highly qualified Lebanese, especially youth, are migrating. With the unemployment rate severely on the rise, an increasing share of households is facing difficulty in accessing basic services, including electricity, health care, water supply/sanitation and education.

Against this background, Lebanon's social and economic woes are being further exacerbated by the negative impact of Russia's war in-Ukraine, which mostly affects: (i) food (in)security, due to possible shortages in imports of crucial foodstuff and agricultural inputs; and (ii) shortages in fuel items due to the global rise in energy prices, and commodity prices more generally, aggravating Lebanon's energy crisis and inflation-depreciation spiral. In addition, negative impact is expected on the action preparedness of the security services (LAF and ISF), increasingly creating security concerns.

The electricity crisis is at the heart of a dysfunctional post-war power-sharing system and has contributed to around 46 % of the Lebanese public debt[2]. The post-war heavily subsidized electricity system has never provided full-time electricity. The Lebanese power stations owned by Electricité du Liban (EDL) run on imported polluting fuel and rely on an outdated transmission and distribution system where only 57 % of electricity produced is actually delivered, billed and collected, as the remainder is lost for technical and non-technical reasons.

[1] WB (2022), Lebanon Economic Monitor, Fall 2021: The Great Denial.

[2] IFI-AUB (2021), Unbundling Lebanon's Electricity Sector.

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EDL's unsustainable model suffers from high production cost, artificially cheap tariffs, low tariff collection rates, poor governance and lack of transparency. Since the summer of 2021, EDL has only produced up to 2-3 hours of electricity per day, forcing households and businesses to increase their reliance on privately owned, expensive and polluting diesel generators. The electricity deals signed with Egypt (for the purchase of natural gas) and Jordan and Syria (for the purchase of Jordanian-produced electricity) are yet to materialize given that the necessary funding backing the deals, to be provided by the World Bank, is conditional on reforms that are not yet being implemented.

The inability of EDL to supply sufficient hours of electricity per day and the lack of any solution in the horizon has led most households and businesses that can afford to invest in Solar PV systems to do so hastily, matter which created a boom in the local market that contributed significantly to the penetration of the renewable energies. The fast development in the demand for Solar PV also caused some inefficiencies in the market as many new contractors started working in solar technology without adequate capacity building which requires more comprehensive approaches in terms of capacity building of the labour force and implementation of the standards of the best practices of the trade.

Globally and as a result of trade and production disruptions caused by the Russian invasion of Ukraine, following the increase of the average price of Brent crude oil to USD 100 a barrel in 2022, its highest level since 2013 and an increase of more than 40% compared with 2021, prices ranged USD 81 a barrel in 2023. In Lebanon, this has pushed energy prices even higher. Diesel (mostly used for generators and transport) which was selling at 331,000 Lebanese lira per gallon - half the Lebanese minimum wage - has already undergone a 450 percent increase since the start of the Russian invasion of Ukraine.

In fact, investment in RE creates numerous job opportunities[3]. In Lebanon, driven by the significant growth of private investment, solar PV value chain has added at least 600 new jobs and the number of the companies has grown from 13 companies in 2010 to 55 companies by the end of 2016[4]. The UNDP study have shown that investment in solar PV system creates various local "direct" and "indirect" jobs in Lebanon[5]. Direct jobs include construction works, the design of the solar PV system, and other business activities such as financing, admission, legal services, consultation and planning. In addition, along the solar PV value chain, the demand for wiring, cabling, legal services etc. creates indirect jobs in the respective economic sectors. It is estimated that each $1 million investment for 1 MW of solar PV installation creates at least 11 direct jobs (Full-Time Employment equivalent, FTE) and 20-25 indirect ones, amounting to 30-45 FTE jobs along the said value chain. This indicates that further acceleration of investment in the installation of solar PV will expand job-demand in the sector. Moreover, it is expected that once the markets for commercial solar thermal, biogas and EE measures (notably heat recovery) are initiated and promoted, new employment opportunities will be created in a similar manner.

Since the current Lebanese energy service providers (EDL and private generator) are unreliable, expensive and dependent on imported fuel, there are untapped market opportunities for RE and EE solutions albeit innovation, entrepreneurship, and technology transfer are necessary to open up these markets. Even though innovation and entrepreneurship have great potential to promote new energy markets, innovation via research and development (R&D) is generally very weak due to insufficient and irregular funding. Strengthening linkages between university research and market/industrial needs can address this innovation challenge. In terms of entrepreneurship, the World Bank identified the most significant success factor for long-term success (hiring employees over time and continuing to do) of Lebanon's startup is additional work experiences[6]. Participation in acceleration programs has no significant effect on long-term hiring probability and serial entrepreneurs in Beirut are less likely to hire consistently. These issues imply that it is very critical to get buy-in from private companies for innovation and entrepreneur program and nurture sector-specific market/industrial insights within human capital for the sector (i.e. entrepreneurs, researchers, students) in order to promote innovation and entrepreneurship. Technology transfer can also make up for lack of R&D funding and available commercialized RE/EE technologies by introducing internationally viable solutions as proven by past UNDP's demonstration projects[7].

[3] IRENA (2018). Renewable Energy and Jobs – Annual Review 2018.

[4] UNDP/DREG (2017). 2016 Solar PV status report for Lebanon.

[5] UNDP (2019). Prioritization and Assessment of Value Chains within the Renewable Energy Sector in Lebanon.

[6] WB (2017). Tech Start-up Ecosystem in Beirut: Findings and Recommendations.

[7] The opportunities in price reduction and increase in PV service providers witnessed from the EU funded CEDRO 4 project.

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Relevance of the Action

This project will provide solar energy solutions to the public sector, thus ensuring the continued provision of essential services and to bridge the energy gap that the national utility is unable to meet now. By doing so, the cost of electricity on the national budget is also lowered which would alleviate the financial burden on both the central government and on the sectors themselves, allowing them to use the financing for other essential services that are needed.

The action supports the private sector in providing technical assistance in this sector to ensure adequate technology transfer, identifying gaps in the market, and liaising with the government to ensure the growth of renewable energy supply in the private sector through innovation and digital solutions. Coupled with training programmes to improve the quality of renewable energy systems provision in the market and policy support to de-risk the sector are means to grow the market and ensure quality systems that are sustainable. The project will provide mentorship for existing SMEs and assist them in identifying potential for scaling up their operations. It will also assist new ventures (start-ups) to reach the marketplace, through mentoring, seed funding, and networking platforms (through a partner incubator).

On a global level, the Action is also aligned with the Lebanese unconditional National Determined Contribution (NDC) under the Paris Agreement to reduce by 20 % the Greenhouse gas emissions (GHG) by 2030, increase the renewable energy to reach 18% of the electricity produced by 2030 and a 3 % energy efficiency target by 2030. It also aligns with the national agenda: the Ministry of Energy and Water's Electricity Policy Paper and Lebanon's Recovery, Reconstruction and Reform Framework (3RF)[8]. The 3RF is a people-centered recovery and reconstruction framework launched by the EU, the World Bank and the United Nations following the 4 August 2020 Beirut Port explosion, to put Lebanon on the path of sustainable development focusing on essential actions, such as addressing the urgent needs of the most vulnerable populations and small businesses affected by the crisis.

In line with the European Commission policy priorities[9], this action will support the external dimension of the European Green Deal and in particular the uptake of sustainable energy solutions, circular economy and green recovery in both the public and private sector. In the private sector, this action will support the industries to transition to renewable energy and adopt circular economy practices and value chains. In the public sector, it will support pressing needs to increase access to affordable electricity and comprehensive energy management solutions (demand-side management, energy efficiency and small-scale renewable systems), and eventually improve the provision of basic public services notably in the justice, education and security sectors.

This action will notably build on the success of CEDRO IV and V to further support innovation and sustainable energy demonstration projects in the industry, relevant regional programmes like Switchmed II, MeetMed or Green Economy Financing Facility. The role of the EU is to promote best practices for sustainable energy and circular economy in industry and public buildings. This approach is based on existing and successful programmes in Lebanon to accelerate the green energy transition while creating sustainable jobs.

[8] The 3RF provides a roadmap to operationalize the finding of the Rapid Damage Needs Assessment (RDNA) and other assessments carried out in response to the devastating port explosion where a large amount of ammonium nitrate caused at least 210 deaths, reported missing persons, and more than 6,500 injured.

[9] EU (2019), The von der Leyen Commission's priorities for 2019-2024.

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II. STRATEGY

The project strategy consists of approaching the transition to renewable energy in Lebanon at two levels: one is to provide and improve access to energy for the public sector, including the Lebanese Armed Forces and the Internal Security Forces and the second is to support the private sector to upscale solar energy supply to greening recovery and creating jobs. Parallel to these two approaches, improving capacities, policies and legislation to push the sector forward will also be tackled based on the lessons learnt and needs identified during implementation at field level of both components. Ultimately, the overall impact of the project is the improved energy security and transition to sustainable energy and resource efficiency in Lebanon. This will be reached based on the below theory of change and in the later sections, more details are provided about the detailed outcomes, outputs and activities.

Theory of Change (TOC)

If (Activities)

- Deployment of sustainable energy solutions for public services (justice, education and security)
- Empowering innovation and entrepreneurial spirit within the sustainable energy domain, especially among the younger demographics and women
- Augmenting the capacity, raising awareness, and providing comprehensive technical assistance to public entities

Then (Outputs)

- Enhanced energy savings and access to electricity via small-scale renewable energy installations
- Reliable energy supply for public service delivery
- Boosted job opportunities and entrepreneurial ventures within the sustainable energy sector
- Upscaled capacity and expertise of industries and public entities to embrace sustainable energy solutions

Then (Outcomes)

- Strengthened energy security and resilience in Lebanon
- Enhanced public service delivery
- Increase in job opportunities and economic growth via the energy sector
- Improved market landscape for renewable energy and sustainable practices in Lebanon

Then (Impact)

- Sustainable and resilient economic development in Lebanon
- Improved quality and reliability in public service delivery
- Balanced gender participation in economic activities

Assumptions (Risk Management)

- Active participation and adoption of sustainable energy practices by industries, SMEs, public entities, and start-ups
- Availability and allocation of ample resources and funding for the project execution
- Stable political and economic landscape facilitating the seamless implementation and progression of the project
- Consistent collaboration and endorsement from local partners and stakeholders

The project's Theory of Change (ToC) aligns with the UN Strategic Development Cooperation Framework (UNSDCF) and the UNDP Country Programme Document (CPD) by providing a definitive pathway to sustainable development, enhancing public services, generating employment, and bolstering energy security, all of which are core constituents of both strategic plans.

The chosen strategy of fostering close alliances with the private sector and public entities is grounded in UNDP's successful track record with analogous initiatives in Lebanon and globally. Existing evidence validates that such a participatory approach amplifies the adoption of sustainable energy solutions, culminating in the desired transformative changes.

Lessons Learnt

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The EU undertook an external midterm review of its renewable energy support programme for Lebanon in the second half of 2022, which includes an analysis of the work of the UNDP-implemented CEDRO5 project. This Action takes into account the lessons learnt from the midterm review of the last EU energy programme (CEDRO V, REFIT and REESTART programmes) and its recommendations for the future of the sector at large. The midterm review recommended several approaches of which some have been incorporated into this Action while others fall outside the scope of UNDP's work. The relevant recommendations are:

- The continuation of support for small scale renewable energy in the private sector in the absence of commercial banking activities and International Financing Institutions (IFIs) and the development of municipal/industrial microgrids and with a focus on smaller industries and agro industries
- The continued technical and financial assistance to small-scale renewable energy in the public sector
- Support the implementation of the Distributed Electricity Law – DEL (if adopted)

Accordingly, in this Action, UNDP will focus on the continued support for small-scale renewable energy solutions for the public sector across various sectors which were prioritised based on needs, impact and importance to the continuation of public service delivery for the most vulnerable and includes security and safety for the Lebanese Armed Forces personnel,the Internal Security Forces, the Justice and Education sector.

Building on the recommendation of the midterm review, and in the continued absence of commercial banking activities both locally and from international partners, this Action will provide technical and financial (albeit limited) assistance to the private sector for innovative renewable energy solutions, including microgrids, through the provision of partial co-fundingthat should be complemented by investment financing from other sources. The Action will explore the possibility for ESCOs to access investment financing from external sources, but to a limited technical extent, given that ESCOs do not have any collateral and are considered high risk for lending within the current financial environment in the country.

Various sustainable energy policy and technical aspects will also be targeted within this Action and will therefore support in advocating for and if adopted, the implementation of the Distributed Electricity Law as well as other related legislation.

Gender

The challenge of achieving sustainable development is intricately linked with the need for gender equality. Unfortunately, a gender gap exists in renewable energy sector in Lebanon, particularly in terms of work opportunities for women in the energy value chain.

Building on UNDP's previous work to bridge the gap in the field of gender and energy as well as climate change action, gender aspects will be mainstreamed across the various activities in this Action. This will include the promotion of women in the solar energy vocational programmes and the encouragement of their active participation in trainings to encourage women technicians and installers in the sector. Gender considerations will also be integrated in the innovation activities to encourage young women entrepreneurs to participate in sustainable energy and climate start-ups and ideas. Women start-ups will be encouraged as well as women-led private sector beneficiaries within this Action.

Furthermore, the project will engage with policymakers, advocating for gender-inclusive energy policies and frameworks. By highlighting the social, economic, and environmental benefits of gender-responsive energy solutions, it aims to create an enabling environment for change.

Ultimately, the project strives to create a future where the involvement of youth and women in the sustainable energy and climate challenge is fully integrated. By championing gender equality in the renewable energy sector, it aims to unlock the potential of women, promote economic empowerment, and pave the way for a more sustainable and equitable world.

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III. RESULTS AND PARTNERSHIPS

Expected Results

Component I. Sustainable Energy for Public Service Entities

Outcome 1: Increase the quality and access to public services through sustainable energy solutions

Public service entities, whether at national or local levels, are struggling to operate due to the low level of electricity supply from the public electricity company Electricité du Liban (EDL) (2-3 hours per day on average). This is affecting the proper operation of schools, hospitals, Ministries, municipalities and all other public service providers all over the country, and therefore impacting the quality and continuity of public service delivery. The increased reliance on fuel generators has caused additional costs that the institutions concerned are unable to cover or pass on to final users, hence leading to major disruptions in the proper functioning of the State and local administrations.

The sharp deterioration in the provision of basic public services due to the electricity crisis have concrete implications for the functioning of schools, hospitals and other basic services leading to partial closure of schools or health care facilities in certain extreme cases. According to the World Bank, the collapse of public service provision is expected to have serious and long-term consequences in terms of human capital loss for the country[10]. Providing the public sector with a more reliable source of electricity is therefore crucial to improve the delivery of essential public services, in particular for the most vulnerable population

- **Lebanese Armed Forces (LAF)**
  LAF is one of the largest institutions in Lebanon with approximately 84,000 active personnel and operates as a conventional military force, although it also plays a significant role in internal security. LAF is a key institution in Lebanon to guarantee the stability of the country and to ensure security in a difficult political and economic climate. Being the most sizable public institution that has earned national status of being both trustworthy and an emblem of Lebanese pride, the LAF is now under operational pressure and strain to continue providing the essential security services to Lebanese citizens. Initially LAF spearheaded the sustainable energy transition within its services, and the investment made is now of utmost usefulness to keep the facilities operational. Energy is a critical component for the LAF's ability to continue their mandate.

  The LAF is also the only public institution in Lebanon to have prepared its own Sustainable Energy Strategy. This strategy was launched in 2017, aiming to increase energy efficiency and renewable energy in its built environment so that the LAF achieves, as an institution, the same Climate Change targets as set by the Government of Lebanon (GoL) under the Paris Agreement through its Nationally Determined Contributions (NDCs). These measures not only lower greenhouse gas emissions (GHGs), but they also provide the LAF with savings on operating costs in a budget-constrained environment. The LAF has provided an energy reprioritization report to the donor community due to the current energy crisis.

  Following the completion of the EU-funded Sustainable Energy for Security SE4S project in 2022 and building on the published guidelines related to the "*Best Practice for Sustainable Energy in The Armed Forces*", "*Sustainable Energy for Security - Hybrid Solar PV, Micro-Wind with Storage*", and mainly the "Sustainable Energy Strategy of the Lebanese Armed Force", the current intervention will focus on providing RE/EE measures to critical and medical facilities part of the LAF.

  The LAF healthcare service facilities, in the form of hospitals and clinics spread across all the regions of Lebanon, support approximately 400,000 Lebanese citizens, consisting of all the military staff required medical care with their direct family members. On the other hand, reliable energy provision is essential to many critical LAF facilities that play a key role in the LAF's mission to protect and serve Lebanon, especially since inadequate energy supply undermines the capacity of these facilities and personnel to effectively carry out their responsibilities.

[10] World Bank, Spring 2021

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This Action will target a minimum of 10 hospitals/health care units and 15 critical facilities (barracks and bases) spread across the country. The prioritization and selection of the facilities was done in coordination with the LAF engineering units.

Examples of some of the renewable energy and energy efficiency interventions that may be implemented could include solar photovoltaic systems with or without battery backup, lighting retrofit and motion detectors to conserve electricity, and solar hot water systems. The exact systems/interventions that will be installed will depend on the energy audits undertaken and the overall budget available for this activity.

- **Internal Security Force (ISF)**

Another related sector to security that has expressed its need for sustainable energy sources is the Internal Security Forces (ISF). Given the ISF's critical role in maintaining urban and rural security amongst citizens, particularly now given the increased tensions because of the financial crisis which has caused additional strain on service provision, the ISF force needs support to maintain its service provision across the country. The ISF also provides citizens with civil registry services, vehicle registration services and other types of civil services which are also essential to continue. Providing sustainable energy solutions to the priority facilities of the ISF for their operation is needed and will be based on the ISF's Solar Study completed in 2022 by ISF engineers. This project will verify the data and prioritise the installations and technical needs accordingly.

Currently, the ISF sites consume yearly more than 7 million litres of diesel to provide proper power supply in their barracks, prison and detention centres and at a cost of more than 5.6M USD (calculated at the price of 800 USD per 1,000 Litre of Diesel). This Action will target a minimum of 20 barracks out of the 227 barracks spread across the country. The prioritization and selection of the barracks was done in coordination with the ISF's engineering units.

The identification of these priority targets are based on discussions held with the ISF team and on the ISF Solar Study of 2022, energy audits will be conducted after which the renewable energy and energy efficiency measures that are most cost-effective will be selected for the prioritised barracks, noting that the number of targeted sites could be increased depending on the availability of budget.

Examples of some of the renewable energy and energy efficiency interventions that may be implemented could include solar photovoltaic systems with or without battery backup, lighting retrofit and motion detectors to conserve electricity, and solar hot water systems. The exact systems/interventions that will be installed will depend on the energy audits undertaken and the overall budget available for this activity.

- **Justice Sector**

The current intervention will focus on the main courthouses across the country and would include the Tripoli, Baabda, Jdeideh, Nabatieh, Saida and Zahle courthouses.

Detailed energy audits and surveys will then be conducted at the courthouses and following the completion of the energy audits, renewable energy and energy efficiency measures will be selected. The priority of interventions will focus on measures that ensure continuous power supply at the courthouses while reducing the cost of diesel to run the generators. Measures directed at improving lighting quality and comfort levels at the facility will also be prioritized based on the assessments and needs of the different buildings.

- **Education Sector**

One of the potential targets of this component is the education sector. In addition to the Ministry of Education's building and the regional education offices, Lebanon has approximately 3,000 education establishments, comprising 1,236 public schools, 97 technical schools, 65 UNRWA supported schools, 331 private yet socially supported schools (e.g., supported by religious establishments), 1,164 private schools, 1 public university that caters to nearly 70,000 students, 36 private universities, and 9 private colleges. In 2021, more than 1 million students were enrolled in the above institutions. All these facilities consume considerable

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amounts of electricity and energy and many of them are seeking donor support with energy efficiency and renewable energy systems to provide the power reliability required to carry out their tasks and to lower the financial burdens of energy provision.

The provision of sustainable energy to schools is vital to carry out in-person learning. This is important not only because it is arguably better education-wise, but also given the electricity and internet availability situation of school and university students at home. In a recent survey of 3261 respondents, more than half the students' parents surveyed indicated that they face 9 to 20 hours power cuts per day, undermining the capacity for online learning[11].

The project would aim to support mostly public but also private sector education institutions, including religious and charity ones where relevant, with sustainable energy provisions. This support will enable better energy reliability to selected education facilities, better budgetary management, and lower environmental footprints. As regards private schools support, an assessement will be performed to ensure that the support is only given to schools whose revenue stream or potential access to funding is insufficient to guarantee the necessary energy supply for its activities costs (details and amounts will be determined at a later stage).

The detailed specifications of interventions, both through energy efficiency and reduction and through renewable energy, will be designed in selected public and NGO-led schools, based on energy audits that will be carried out by UNDP once the project is initiated. The solar equipment and interventions will then be prioritised based on the results of the audit and those that have low payback periods and are most cost-effective will be implemented.

Examples of some of the renewable energy and energy efficiency interventions that may be implemented could include solar photovoltaic systems with or without battery backup, lighting retrofit and motion detectors to conserve electricity, and solar hot water systems. The exact systems/interventions that will be installed will depend on the energy audits undertaken and the overall budget available for this activity.

The mapping phase for the education sector is an ongoing process that requires extensive coordination with the focal points at MEHE and DGVTE, inter-UN family coordination (in particular with UNICEF) and with the different NGOs and entities working on improving the sector's conditions. First, a detailed listing for all types of schools in Lebanon will be developed as per the below:

- Public/Private/NGO-led Institutions
- Schools/Technical Schools
- Location: Governorate/District/Town/Village/Coordinates
- Number of Students
- 1 shift for Lebanese students only or 2 shifts for Lebanese and Syrian students

Following the first phase of the assessment and after reaching a complete compiled list of institutions, a survey will be shared with the schools to gather details related to their existing conditions and as per the below:

- Facility Details:
    - Total Built Area (m^2)
    - Total Number of Buildings
    - EDL Meter Size
    - Generators with Capacities and/or Generator Subscription Details
- Energy Details:
    - EDL Bills
    - Generator Bills
- Possible Interventions/Donors:
    - Previous energy-related projects details (Donor, System with Details)

[11] Center of Lebanese Studies, 2021

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- Future energy-related project details (Donor, System with Details)

After completing the survey phase, the below selection criteria for the institutions in need of assistance will be used:

- School building owned by MEHE, local municipality or public entity
- Absence of RE/EE measures or obsolete/incomplete RE system in place
- Number of students attending the school

The list of schools needing interventions will then be finalized in line with the above and in coordination with the different sectors leads in MEHE and DGVTE. Detailed site visits will then be conducted to gather the needed data and details, after which energy audits will be initiated and the ensuing RE/EE measures will be selected accordingly.

- **Public Services Facilities:**
  There are other public service facilities at local level that are also critical to ensure the continuation of civil sector public service provision and the interventions can expand to provide RE/EE interventions to 14 Serails spread across the country, and which would assist in ensuring better access to public services for residents. These Serails are public building that fall within the jurisdiction of the Ministry of Interior and Municipalities and house offices from different public sector departments including as examples, ISF sub-regional offices, fire brigades, etc.

- **Capacity Building & Vocational Trainings**
  Capacity building will be undertaken at two levels: one is at the level of each facility where the renewable energy systems will be installed and another level will be the capacity building of technicians and the public on sustainable energy in general, technologies and climate change.

  At the first level, trainings will include the provision of energy management certification(s) to the facility staff of the above-mentioned public sector facilities part and on energy interventions implemented so that they can maintain and operate the installed systems properly after the intervention.

  At the second level, to improve the quality of the overall working environment, the Action will focus on improving the quality of technicians and renewable energy service providers through the creation of vocational programmes and the training of experts in the field. The Action will strengthen the certification of the labour force and technicians on best practices for the design, installation, testing and commissioning of renewable energy systems and energy efficiency measures. .

  In addition and to complement the activites implemented by UDNP under these 2 specific levels, UNDP will improve through this Action and others the sustainable energy value chain from design, installation, material import quality and standards to commissioning and certification.

Summary of Component 1 and its related Outputs and Activities as per the logframe

Outcome 1: Increase the quality and access to public services through sustainable energy solutions

This outcome will focus on enhancing the capacity of the Lebanese Armed Forces,the Internal Security Forces Education and Justice ministries to perform their duties through the implementation of their respective sustainable energy strategies.

Outputs Contributing to Outcome 1

1.1 Increased energy savings and access to electricity from small scale renewable energy installations in public service facilities
1.2 Increased awareness and capacity on Sustainable energy management and climate change

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Activity Output 1.1

- Identification of priority facilities and procurement of technical consultants (as needed)
- Energy audit reports of selected key facilities and identification of priority RE interventions
- Solar photovoltaic and/or other renewable energy systems interventions
- Energy lighting retrofits and control interventions
- Solar thermal systems interventions
- Heating, ventilation, and cooling interventions
- Computerized maintenance and monitoring systems interventions

Activity Output 1.2

- Capacity building
    - Trainings to engineering staff on design, installation, and commissioning of interventions
    - Trainings on operation and maintenance of project interventions
    - Trainings on sustainable procurement
- Vocational trainings
    - Certification of Level 1 and Level 2 Technicians for Solar PV and SWH installation, commissioning, performance testing and operation and maintenance.
    - Developing curriculum for vocational trainings in diverse specializations and adapting them to the market needs.
    - Implementation of vocational trainings
    - Advocacy to have the trainings accredited to promote a TVET system that provides the competencies and skills required in the labour market.
    - Sustainability of the project through the upgrading of the current curricula in formal technical education of 3 BT level specialties.
- Internship programmes at UNDP
- Policy papers and technical analysis as needed
- Material development and dissemination
- Awareness raising activities and EU visibility on sustainable energy

Component II. Sustainable Energy for Job Creation and Economic Development

Outcome 2 : increase the competitiveness of the industrial sector through sustainable energy, resource efficiency and circular business models and enhance a more equitable and gender-balanced participation of all stakeholders

Industry is one of the major sectors in Lebanon with potential to drive economic recovery, local employment, and the generation of hard currency that is urgently needed. However, high costs of energy have had a direct impact on inflating the operational costs of industries, decreasing their competitiveness and even their ability to sustain operations given the current economic crisis in Lebanon. The high cost of energy reduces the chances of much needed investments such as research, development, and upscaling operations that could generate employment opportunities.

UNICEF has indicated that 31% of young people in Lebanon are not in education, employment or training as of 2021.[12] The Lebanese economy is composed by 95% of Small and Medium Enterprises (SMEs) that represent 50% of total employment.[13] The industrial sector employs 23.6% of the Lebanese workforce (2019 data), and thereby enabling the sector to retain and grow its workforce is essential to economic wellbeing. Reducing energy use also has an indirect economic impact as every Euro saved from purchasing imported fuel is a Euro invested in the local economy with multiplier effects.

The objective of this outcome is the increased competitiveness of industry through sustainable energy, resource efficiency and circular business models and enhance a more equitable and gender-balanced participation of all stakeholders. The objective will be met by working mostly on two approaches with the private sector: one relates to

[12] UNICEF Youth-Focused Rapid Assessment (YFRA) 2022. https://www.unicef.org/lebanon/media/7746/file

[13] https://www.economy.gov.lb/media/11222/smes-in-lebanon-180412-19-website.pdf

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strengthening innovation and entrepreneurship and the second relates to technical assistance on renewable energy approaches to the industrial, commercial and private business sectors.

- **Private Sector Technical Assistance**
  As identified by both IRENA[2] and the UNDP[4], the renewable energy and energy efficiency measures showed a promising potential for local job creation, which further strengthen the choice of their implementation especially that it assists Lebanon in reaching their targets and it also assists the project in creating new job opportunities. At the same time, based on UNDP's previous experience in the field with industries in specific, there is a technical knowledge gap in most business processes about the energy savings potential that renewables and energy efficiency measures can provide. While for those facilities that are aware, financing challenges including access to funds and technical support are gaps.

  At the same time, pilot installations within the private sector SMEs, industrial and commercial facilities provide opportunities for innovative approaches to sustainable energy solutions that would upscale renewable energy solutions and pilot digitalisation. By providing technical assistance and potentially grant co-financing to industries of up to 15% and at a cap of $600,000 of their innovative investment project, new types of renewable energy approaches will be tested, thus expanding the market and providing lessons learnt for others. Beneficiaries will be selected on a competitive basis and specific technical selection criteria will be used (these will be developed during the inception phase of this Action). Performance-based contracts will be signed with beneficiaries and the partnership will delineate the roles and responsibilities of each party.

  UNDP will also coordinate with investment financing institutions, , to support on renewable energy financing to the private sector. The project will also coordinate with relevant industry representativies to network with all SMEs and industries on this Action and for knowledge transfer.

- **Innovation and Entrepreneurship**
  The strategy within this current Action is to continue with the same approach that was started in the EU-funded CEDRO5 project by pushing forward on innovation through energy innovation hubs in Lebanon while strengthening the market for green energy jobs through online energy platforms such as the "Energy Hub", operationalising green building certificate programmes and approaches, and strengthening the renewable energy certification programme in Lebanon.

  Currently, the established "Energy Innovation Hub" (EIH) works as a one-stop innovation and knowledge platform / meeting point for the EE and RE sectors to accelerate multi-stakeholder collaborations via the creation of shared tools and resources for SMEs, deployment of new business models and market solutions by bringing E-I-SMEs and research organizations together.

  The project will work on the EIH platform aiming to:

  1. Ensure the financial sustainability of the platform by generating sufficient income streams to cover its expenses. Such income streams shall be set-up to the extent possible in an automated manner
  2. Ensure the viability of the platform by generating development impact and synergies in the energy market. Each impact shall have a specific growth strategy and a monitoring dashboard on the platform
  3. Incentivize different energy users and stakeholders in the field to sign-in and use the platform.

  The Action will also continue to strengthen the building certification programme " Arz[14]" in terms of marketing and strengthening its application process while linking it to other on-going building certification programmes in the country.

  Lebanon's start-ups have proven their ability to address the country's biggest problems when it comes to export, transportation, energy, and agriculture. They have proven resilient to instability and even acted as a major driver not only in saving jobs but also in creating them. Therefore, it is highly imperative that the journey

[14] https://www.arzrating.com/en

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of EIH continues and imprint its impact and success by supporting yet another promising cohort of Lebanese energy entrepreneurs and guide them into creating and improving innovative energy saving and efficient solutions tailored to the country's energy industry. After the successful conclusion of the first EIH program cycle cohort which incubated and accelerated 23 Lebanese energy start-ups, UNDP, through arelevant organisation to be selected according to UNDP procurement rules, will support the development of a new batch of EIH (EIH 2.0) as it's imperative to the program's track record that the efforts continue to bring forth a new wave of promising energy start-ups in Lebanon. The new EH 2.0 will be built upon the success of the 1st EH and the lessons learned concluded in the EIH roadmap.

UNDP, through **Nucleas Venture** as an Implementing Partner for this activity, will promote innovation and entrepreneurship in support to Lebanon's clean energy transition, through synergies created with the main players in the energy sector to bridge the existing gap between the R&D sector (green entrepreneurs, innovation promoters, universities and research organizations) and local industries and SMEs. The project will launch a series of call for ideas for new ventures in support of the energy sector and will provide the required support as identified by the MOET's strategy to enable green economy growth and job creation. The project will also provide support to NV to properly identify start-ups and innovative business models that answer local market needs. In addition to keeping up with the core program pillars introduced in the first edition, the successful completion of a second edition will also rely on maintaining and improving two important program offerings: access to labs for product testing and development and focused one on one mentorship for the start-ups: access to labs, focused mentorship, enhancing ideation stage, host an Idea Validation Bootcamp, Start-up Cohort, etc.

Finally, setting up the system of Renewable Energy Certificates (RECs) for Lebanon, even on the envisioned voluntary basis, is an innovation that has the potential to significantly expand the renewable energy market in Lebanon. Commonly, REC prices range from 5 – 10 Euros per REC (each REC is equal to 1 MWh). After setting up the REC system in Lebanon legally and technically, the reach out of solar farm owners to the Lebanese Centre for Energy Conservation (LCEC) is still considered low in terms of applications received. In addition to that, LCEC are still in need for technical support in receiving and reviewing applications to assure that a smooth process in issuing RECs. UNDP, in coordination with relevant sector organisations, will support in strengthening the RECs market in Lebanon through supporting solar farm owners and LCEC. UNDP, in coordination with **Stichting I-REC** (EU based NGO), U will be piloting the RECs generation through supporting several solar farms to register with LCEC and generate RECs. Different methods will be piloted including claiming RECs and selling RECs. In addition to that, several workshop (in-person and virtual) will be hosted targeting energy stakeholders in Lebanon, investors, ESCOs, RE companies, solar farm owners, etc. to elaborate more on the importance of RECs and the registration method. This will help in creating a RECs market in Lebanon and will stabilize the REC price according to the market's supply and demand. In addition to that, the project will support LCEC, the local REC Issuer in Lebanon, by training their RECs focal points to accept applications, review them, and issue RECs; this will also help in rolling this market in Lebanon.

- **Technical studies and awareness raising**
  In addition, technical, legal and policy studies will be prepared, in close coordination with the Ministry of Energy and Water and LCEC, to promote the upscaling of sustainable energy and energy efficiency based on UNDP and the partners' experience from the ground. Some of the analysis could include an update of the Derisking Renewable Energy Interventions, Sustainable Energy Strategy for Industries, and others.

Summary of Component II and its related Outputs and Activities as per the logframe

For Outcome 2 Sustainable Energy for Job Creation and Economic Development

To increase the competitiveness of the industrial sector through sustainable energy, resource efficiency and circular business models and enhance a more equitable and gender-balanced participation of all stakeholders

Outputs Contributing to Outcome 2

2.1Increased energy savings and access to electricity from small scale renewable energy installations in industrial and commercial facilities

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2.2Increased job created and entrepreneurs in the sustainable energy field (in particular women)

2.3Increased capacity and knowledge of the industries to adopt sustainable energy solutions

Activity Output 2.1

- Competitive process to select beneficiaries and/or the identification of innovative solutions (microgrid or other) and sign agreements
- Energy audit reports of selected private sector facilities and/or other needed feasibility studies
- SMEs supported technically for the supply and installation of RE and EE interventions (technical assistance and limited grant for innovative projecst)

Activity Output 2.2

- Operational "Energy Hub" platform through incubator serving as a 1-stop shop for the energy stakeholders in Lebanon
- An incubation and acceleration program for young innovators in sustainable energy
- Support to successful start-ups in EE/RE
- E-library for energy incubation program material
- Operationalisation of the RECs programme in Lebanon

Activity Output 2.3

- Advocacy meetings and discussions to promote sustainable energy and energy efficiency measures
- Legal, policy and technical recommendations through studies and analysis including but not limited to:
    - Sustainable Energy Strategy for the Lebanese Industrial Sector
    - Update the "De-risking Renewable Energy Investment" for Lebanon

Resources Required to Achieve the Expected Results

This project will be a multi-donor action funded by EU and UNDP. To ensure effectiveness of programme delivery, the action will benefit from the dedicated project team, established in a cost-effective manner, whose positions match the stated outputs of the action while maintaining UNDP's quality assurance standards, subject to possible changes as needed to accompany the evolution of national needs and priorities.

An annual monitoring and evaluation plan, including risks and mitigating measures, will be developed based on the annual work plan designed by the project and the UNDP quality assurance team. The project is part of the overall country programme quality assurance structures, which include a solid results-based monitoring system and data collection and monitoring capacity. In addition, and based on specific punctual needs, individual consultants (experts), consulting companies and contracting companies will be procured based on UNDP procurement guidelines to fulfil and compliment specific capacity needs.

Partnerships

Public sector stakeholders are primarily the Ministry of Energy and Water, Ministry of Health, Ministry of Industry, Ministry of Education and Higher Education, Ministry of Environment, Ministry of Interior and Municipalities, the Ministry of Justice, the Lebanese Armed Forces and the Internal Security Forces.

Furthermore, the project will also coordinate and work closely with the Lebanese Centre for Energy Conservation, the Order of Engineers and Architects, the Lebanese Green Building Council, the Lebanese Solar Energy Society and other NGOs and civil society associations working in the sector of sustainable energy and climate change. These stakeholders will be engaged by the UNDP team to coordinate activities, ensure policy dialogue, exchange knowledge and information but without any financial payments.

Risks and Assumptions

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Lebanon's political, economic and security situation remains unstable. This situation may last the entire duration of the project implementation period. Mitigation measures are included for each of the risks mentioned in the table below and will be continuously assessed and updated by UNDP. Any changing or criticalrisksk will also be escalated to the Project Board as needed.

Table 1. Project Risks and Mitigation

Category	Risk	Likelihood (High/Medium/ Low)	Impact (High/Medium/ Low)	Mitigating measures
External environment	Damage to installed equipment and material	Medium	Intermediate	Memorandums of understanding will be signed with beneficiaries to ensure the protection, operation and maintenance of the equipment installed.
	Natural disasters and environmental degradation	Low	High	Ensure all outdoor equipment can withstand severe weather conditions expected in Lebanon. Ensure that an environmental statement is undertaken for all projects with significant installations.
Planning, processes and systems	Battery disposal	High	Medium	UNDP will be working on battery disposal through the EU-funded TaDWIR project. Activities will be coordinated with this Action.
	Ability to effectively operate and maintain installed systems	High	High	A minimum of two-years performance guarantee for all systems installed will be required. The possibility of including O&M contract will also be assessed and capacity building activities will be ensured.
	Delays in import clearance will cause delays to the project	Medium	Medium	Sufficient time will be given for project implementation to account for expected delays in manufacturing, shipping and clearance of imported system components. Continuous coordination will be conducted with the contractors and suppliers to overcome any possible delays in manufacturing or shipping.
	Delays in manufacturing and shipping of equipment	High	High	
People and the organisation	Instability in the security situation in the country	Medium	Medium	Isolated security incidences may occur. Appropriate measures will be taken accordingly. These could include the postponement of some of the works on site, and the coordination with the Internal Security Forces and UN Department of Security as needed.
Legality and regularity aspects	Availability of funds to assist in co-financing private sector contribution	Medium	High	Available funds will be sought, such as CEDAR Oxygen, and support from other institutions (e.g. Cross Boundary)
	Capital controls and the inability to raise bank guarantees	High	High	Contractual agreements to be signed between the Implementing partners and the private sector beneficiaries that would overcome banking challenges to the extent possible.

Stakeholder Engagement

The key stakeholders that will act as partners have been identified and their roles and contribution to the project are described above. The target groups of the project are displaced Syrians and host communities where tensions are high. The engagement of these target groups is ensured by formulating the project based on the extensive consultation with the municipalities, local communities and concerned ministries/institutions.

South-South and Triangular Cooperation (SSC/TrC)

Not Applicable.

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Digital Solutions[15]

Under Outcome 2, pilot installations within the private sector SMEs, industrial and commercial facilities provide opportunities for innovative approaches to sustainable energy solutions that would upscale renewable energy solutions and pilot digitalisation. By providing technical assistance and potentially grant co-financing to industries of up to 15% and at a cap of $600,000 of their innovative investment project, new types of renewable energy approaches will be tested, thus expanding the market and providing lessons learnt for others.

Knowledge

The Action will work extensively on capacity building and knowledge products related to sustainable energy and energy efficiency and conservation. These products will be prepared by in-house experts as well as international and local consultants and will cover various technical, policy and legislative topics in the field depending on what is most needed to push forward the climate mitigation agenda. All products will be posted for public use on the UNDP website.

Furthermore, the Action will work on knowledge building at technical level for the design, installation and operation and maintenance of renewable energy systems and energy efficiency measures particular with the engineering teams of the beneficiary sites. Targeted trainings will be conducted for these personnel to build their capacity.

Sustainability and Scaling Up

The Action builds on nearly two decades of experience in the sustainable energy sector for UNDP and on existing synergies with relevant stakeholders that is on-going through the current CEDRO5 project. The Action will be implemented in close coordination with the Ministry of Energy and Water as the main government focal point on energy to ensure sustainability of the Action, particularly in terms of policy support and technical work.

[15] *Please see the Guideline "Embedding Digital in Project Design".*

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IV. PROJECT MANAGEMENT

Cost Efficiency and Effectiveness

The project uses a portfolio management approach to improve cost-effectiveness and efficiency through synergies with other interventions through the CEDRO Renewable Energy project as well as the parallel renewable energy project funded by KfW. For any balance at the end of the project, the country office will consult with the EU.

Project Management

The Project Team will be responsible for the day-to-day management of the project. The work will be integrated within and managed by the CEDRO project team. The UNDP team will implement and monitor the progress towards intended results and will ensure high-quality managerial, technical and financial implementation of the project, including the activities undertaken by partner entities. UNDP will fully engage with the public and private sector beneficiaries for the implementation of the activities while also coordinating closely with national counterparts such as Ministry of Energy and Water and Ministry of Environment. Given UNDP's historic and on-going presence in various ministries, the technical work can also be disseminated extensively to a wide scope of national partners. This approach ensures the adoption of the project policy recommendations and that all technical knowledge is absorbed at the government level and mainstreaming of knowledge is enhanced at various ministerial levels given the cross-cutting nature of the project in terms of job creation, climate impact, benefit to industries and innovation

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V. RESULTS FRAMEWORK[16]

Intended Outcome as stated in the UNSDCF/Country Programme Results and Resource Framework:
3.1. Environmental Governance Improved.

Outcome indicators as stated in the Country Programme [or Regional] Results and Resources Framework, including baseline and targets:
CPD Outcome 3: Strengthened stabilization and green recovery to reduce vulnerabilities and environmental risks, including through enhanced competitiveness and business environment of sustainability-oriented MSMEs and high potential green productive sector values chains.
CPD output 3.2 (IRRF output 5.2): Transition to renewable energy accelerated capitalizing on technological gains, clean energy innovations and new financing mechanisms to support green recovery.
CPD output indicator 3.2.1.: Amount of renewable energy installed annually (output indicator for UNSDCF output 1.2, IRRF 5.2.1. Increase (in megawatt) in installed renewable energy capacity with solar, wind, biomass and other technologies)

Applicable Output(s) from the UNDP Strategic Plan:
1.5.1. Solutions adopted to achieve universal access to clean, affordable and sustainable energy
2.51. Solutions developed, financed and applied at scale for energy efficiency transformation to clean energy and zero carbon development for poverty eradication and structural transformation

Project title and Quantum Project Number: EU Support to Lebanon's Green Agenda (Sustainable Energy) - 01001256

EXPECTED OUTPUTS	OUTPUT INDICATORS[17]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)					DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	FINAL	
Impact: Improved energy security and transition to sustainable energy and resource efficiency in Lebanon	**1.** Amount of fossil fuel import (Tons/year)	UNDP, MoEW, MoE, CAS, Beneficiries	TBD	2023	0	1,500	3,500	4,900	4,900	Baseline and endline surveys
	2. Greenhouse Gas (GHG) emissions avoided (tonnes CO2eq/year) with EU support		TBD	2023	0	4,000	8,000	12,000	12,000	Baseline and endline surveys
Outcome 1: Increased quality and	**1.1.** Fuel consumption for electricity generation in public service facilities	UNDP, Beneficiries	TBD	2023	0	10%	20%	30%	30%	Experience and data from previously implemented UNDP

[16] UNDP publishes its project information (indicators, baselines, targets and results) to meet the International Aid Transparency Initiative (IATI) standards. Make sure that indicators are S.M.A.R.T. (Specific, Measurable, Attainable, Relevant and Time-bound), provide accurate baselines and targets underpinned by reliable evidence and data, and avoid acronyms so that external audience clearly understand the results of the project.

[17] It is recommended that projects use output indicators from the Strategic Plan IRRF, as relevant, in addition to project-specific results indicators. Indicators should be disaggregated by sex or for other targeted groups where relevant.

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EXPECTED OUTPUTS	OUTPUT INDICATORS[17]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)					DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	FINAL	
access to public services through sustainable energy solutions										projects
	1.2. Number of hours of electricity in public service facilities enabled by renewable energy with EU support		TBD	2023	0	3	4	5	5	Baseline and endline surveys
	1.3. Number of beneficiaries with increased access to energy services		TBD	2023	0	1,000	2,000	3,000	3,000	
Output 1.1. Increased energy savings & access to electricity from small scale renewable energy installations in public service facilities	**1.1.1.** Tons of CO2 emissions saved per year with EU support	UNDP, MoEW, MoE, Beneficiries	TBD	2023	0	2,000	4,000	6,000	6,000	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	1.1.2. Energy saved per year (MWh) with EU Support		TBD	2023	0	1,500	3,000	4,250	4,250	Baseline and endline surveys
	1.1.3. Renewable energy power generated per year (MWh) with EU Support		TBD	*2023*	0	1,500	3,000	4,250	4,250	Baseline and endline surveys
Output 1.2. Increased awareness and capacity on Sustainable energy management and climate change	**1.2.1** Number of trainers trained		0	2023	0	10	20	30	30	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	1.2.2 Number of trained/educated beneficiaries on sustainable solutions and climate change		0	2023	0	500	1,500	2,000	2,000	Based on awareness raising activities of Project within selected beneficiaries
	1.2.3 Number of beneficiaries with knowledge in sustainable solution and climate change		TBD	2023	0	10	20	30	30	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	1.2.4 Number of courses developed (sustainable energy and climate change)		TBD	2023	0	1	2	3	3	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
Outcome 2. Increased competitiveness of the industrial sector through sustainable energy, resource efficiency and circular business models and enhance a more	**2.1.** Number of Micro Small Medium Entreprises (MSMEs) reporting increased turnover from Circular conomy activities as a direct result of EU support received, disaggregated by sex and age group of the owner, enterprise size	UNDP, MSMEs, Beneficiaries	TBD	2023	0	5	15	20	20	Baseline and endline surveys

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EXPECTED OUTPUTS	OUTPUT INDICATORS[17]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)					DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	FINAL	
equitable and genderbalanced participation of all stakeholders	**2.2.** Percent of savings generated from improved resource (material, energy, water) efficiency		TBD	2023	0%	10%	15%	20%	20%	Baseline and endline surveys
Output 2.1. Increased job created and entrepreneurs in the sustainable energy field (in particular women) and Increased capacity and knowledge of the industries to adopt sustainable energy solutions	2.1.1 Number of successful start-ups with EU support		TBD	2023	0	3	5	0	0	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.1.2 Number of women engineers empowered with EU support		TBD	2023	5	10	20	30	30	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.1.3 Number of green jobs supported/sustained by the EU		*TBD*	2023	500	2,000	3,500	5,000	5,000	Baseline and endline surveys
	2.1.4 Number of workshops implemented		0	2023	0	4	8	12	12	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.1.5 Number of trainers trained		0	2023	0	10	20	30	30	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.1.6 Number of industrialists with knowledge in sustainable solution and climate change		TBD	2023	5	20	40	50	50	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
Output 2.2. Increased energy savings & access to electricity from small scale renewable energy installations in industrial & commercial facilities	2.2.1 Tons of CO2 emissions saved per year witih EU support	UNDP, MSMEs, Beneficiaries	TBD	2023	0	2,000	4,000	6,000	6,000	Report EU-funded intervention's M&E system, Progress reports, reviews, evaluations
	2.2.2 Energy saved per year (MWh) with EU Support		TBD	2023	0	1,500	3,000	4,250	4,250	Baseline and endline surveys
	2.2.3 Renewable energy power generated per year (MWh) with EU Support		TBD	2023	0	1,500	3,000	4,250	4,250	Baseline and endline surveys

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VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans:

Monitoring Plan

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analysed to assess the progress of the project in achieving the agreed outputs.	Quarterly, or in the frequency required for each indicator.	Slower than expected progress will be addressed by project management.	-	Internal/no cost
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Quarterly	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.	-	Internal/no cost
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.	-	Internal/no cost
Annual Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually/Biennially	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.	-	Internal/no cost
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.	-	Internal/no cost
Project Report	A progress report will be presented to the Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures, and any evaluation or review reports prepared over the period.	Annually, and at the end of the project (final report)		-	Internal/no cost
Project Review (Project Board)	The project's governance mechanism (i.e., project board) will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project	At least annually	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed	-	Internal/no cost

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Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
	Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.		to address the issues identified.		

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VII. MULTI-YEAR WORK PLAN [18][19]

Responsible Party is UNDP for all activities except for the implementation of the EIH and Innovation activities which will be implemented by Nucleus Ventures for approximately $ 1 million (output 2.1)

Expected Outputs	PLANNED ACTIVITIES			Planned Budget (USD)					
	Activities and budget description	Funding Source	Quantum account	2023 (01 November)	2024	2025	2026	2027 (31 October)	Total Budget
Output 1.1	**A1 - Human Resources**								
	Contractual Services Individual (NPSA)	EU	71400	12,324.00	310,354.00	808,592.00	808,592.00	796,268.00	2,736,130.00
	Salary Cost - NP Staff		61100		129,840.00	129,840.00	129,840.00	129,840.00	519,360.00
	Salary Cost GS Staff		61200		72,036.00	72,036.00	72,036.00	72,036.00	288,144.00
	F&A		75100	862.68	35,856.10	70,732.76	70,732.76	69,870.08	248,054.38
	Subtotal A1			**13,186.68**	**548,086.10**	**1,081,200.76**	**1,081,200.76**	**1,068,014.08**	**3,791,688.38**
	A2 - Office and Travel Expenses								
	Rental and Maintenance Premises	EU	73100		33,153.00	74,151.00	59,698.00	59,698.00	226,700.00
	Rental and Maintenance Premises	UNDP	73100		5,247.00	19,449.00	33,902.00	33,902.00	92,500.00
	Communication and Audio-visual Equipment	EU	72400		4,968.00	4,968.00	4,968.00	4,968.00	19,872.00
	Information Technology Equipment		72800		30,000.00	9,000.00	-	-	39,000.00
	Office supplies		72500		1,440.00	1,440.00	1,440.00	1,440.00	5,760.00
	Travel		71600		20,000.00	20,000.00	20,000.00	20,000.00	80,000.00
	Equipment and Furniture		72200						

[18] Cost definitions and classifications for programme and development effectiveness costs to be charged to the project are defined in the Executive Board decision DP/2010/32

[19] Changes to a project budget affecting the scope (outputs), completion date, or total estimated project costs require a formal budget revision that must be signed by the project board. In other cases, the UNDP programme manager alone may sign the revision provided the other signatories have no objection. This procedure may be applied for example when the purpose of the revision is only to re-phase activities among years.

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					45,000.00	-	-	-	45,000.00
	Materials and Goods		72300		7,200.00	7,200.00	7,200.00	7,200.00	28,800.00
	Rental and Maintenance of Other Equipment		77600		21,600.00	21,600.00	21,600.00	21,600.00	86,400.00
	F&A		75100	-	11,435.27	9,685.13	8,043.42	8,043.42	37,207.24
	F&A	UNDP	75100	-	419.76	1,555.92	2,712.16	2,712.16	7,400.00
	Subtotal A2			**-**	**180,463.03**	**169,049.05**	**159,563.58**	**159,563.58**	**668,639.24**
	A3 - Lebanese Armed Forces								
	Contractual Services Companies	EU	72100	-	100,000.00	819,600.00	1,639,200.00	1,639,200.00	4,198,000.00
	F&A		75100	-	7,000.00	57,372.00	114,744.00	114,744.00	293,860.00
	Subtotal A3			**-**	**107,000.00**	**876,972.00**	**1,753,944.00**	**1,753,944.00**	**4,491,860.00**
	A4 - Internal Security Forces								
	Contractual Services Companies	EU	72100		100,000.00	720,000.00	1,440,000.00	1,440,000.00	3,700,000.00
	F&A		75100	-	7,000.00	50,400.00	100,800.00	100,800.00	259,000.00
	Subtotal A4			**-**	**107,000.00**	**770,400.00**	**1,540,800.00**	**1,540,800.00**	**3,959,000.00**
	A5 - Justice								
	Contractual Services Companies	EU	72100		100,000.00	540,000.00	1,080,000.00	1,080,000.00	2,800,000.00
	F&A		75100	-	7,000.00	37,800.00	75,600.00	75,600.00	196,000.00
	Subtotal A5			**-**	**107,000.00**	**577,800.00**	**1,155,600.00**	**1,155,600.00**	**2,996,000.00**
	A6 - Education & Serails & other								
	Contractual Services Companies	EU	72100		50,000.00	560,000.00	1,120,000.00	1,120,000.00	2,850,000.00
	F&A		75100	-	3,500.00	39,200.00	78,400.00	78,400.00	199,500.00
	Subtotal A6			**-**	**53,500.00**	**599,200.00**	**1,198,400.00**	**1,198,400.00**	**3,049,500.00**

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	A7 - Capacity building, Policy & Communication								
	Contractual Services Companies		72100	-	46,428.00	251,683.00	638,492.00	648,529.00	1,585,132.00
	Interns		71400	-	12,000.00	12,000.00	12,000.00	12,000.00	48,000.00
	Local consultants		71300	-	50,000.00	50,000.00	50,000.00	50,000.00	200,000.00
Output 1.2	International consultants	EU	71200	-	50,000.00	50,000.00	50,000.00	90,000.00	240,000.00
	Trainings, Workshops, and Conferences		75700	-	-	50,000.00	50,000.00	-	100,000.00
	Audio-visual and Printing Production Costs		74200	-	-	23,695.00	23,695.00	23,695.00	71,085.00
	F&A		75100	-	11,090.15	30,616.46	57,693.09	57,695.68	157,095.38
	Subtotal A7			**-**	**169,518.15**	**467,994.46**	**881,880.09**	**881,919.68**	**2,401,312.38**
	A8 - Innovation and Entrepreneurship								
Output 2.1	Contractual Services Companies	EU	72100	-	50,000.00	150,000.00	300,000.00	300,000.00	800,000.00
	F&A		75100	-	3,500.00	10,500.00	21,000.00	21,000.00	56,000.00
	Subtotal A8			**-**	**53,500.00**	**160,500.00**	**321,000.00**	**321,000.00**	**856,000.00**
	A9 - Private Sector Technical Assistance								
Output 2.2	Contractual Services Companies	EU	72100	-	50,000.00	384,000.00	768,000.00	768,000.00	1,970,000.00
	F&A		75100	-	3,500.00	26,880.00	53,760.00	53,760.00	137,900.00
	Subtotal A9			**-**	**53,500.00**	**410,880.00**	**821,760.00**	**821,760.00**	**2,107,900.00**
	Project Total			**13,186.68**	**1,379,567.28**	**5,113,996.27**	**8,914,148.43**	**8,901,001.34**	**24,321,900.00**

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VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Implementation Modality

This project will be implemented under the Country Programme Document of UNDP for 2023 – 2025 and falls within Outcome 3 that targets "Strengthened stabilization and green recovery to reduce vulnerabilities and environmental risks, including through enhanced competitiveness and business environment of sustainability-oriented MSMEs and high potential green productive sector values chains" and will use UNDP's standard implementation modality for Lebanon (Direct Implementation Modality – DIM). The project implementation scope will be between those of various ministries, such as: the Ministry of Energy and Water for the development of renewable energy; the Ministry of Environment for climate change mitigation action; the Ministry of Economy and Trade for job creation and SME support; the Ministry of Industry for the work with industries and with the renewable energy service providers/suppliers; the Ministry of Interior and Municipalities for municipality support and the Internal Security Forces, the Ministry Education, the Lebanese Armed Forces. Thus, it must therefore remain equidistant from all the national stakeholders and for all reasons mentioned above, DIM is considered as the most appropriate mechanism for the Project implementation.

Governance and Overall Project Management

The UNDP will monitor the progress towards intended results, and will ensure high-quality managerial, technical and financial implementation of the project, and will be responsible for monitoring and ensuring proper use of administrated funds to the assigned activities, timely reporting of implementation progress as well as undertaking of mandatory and non-mandatory evaluations for each of their respective components. Furthermore, the procurement of goods and services and the recruitment of personnel shall be provided in accordance with UNDP guidelines, procedures and regulations.

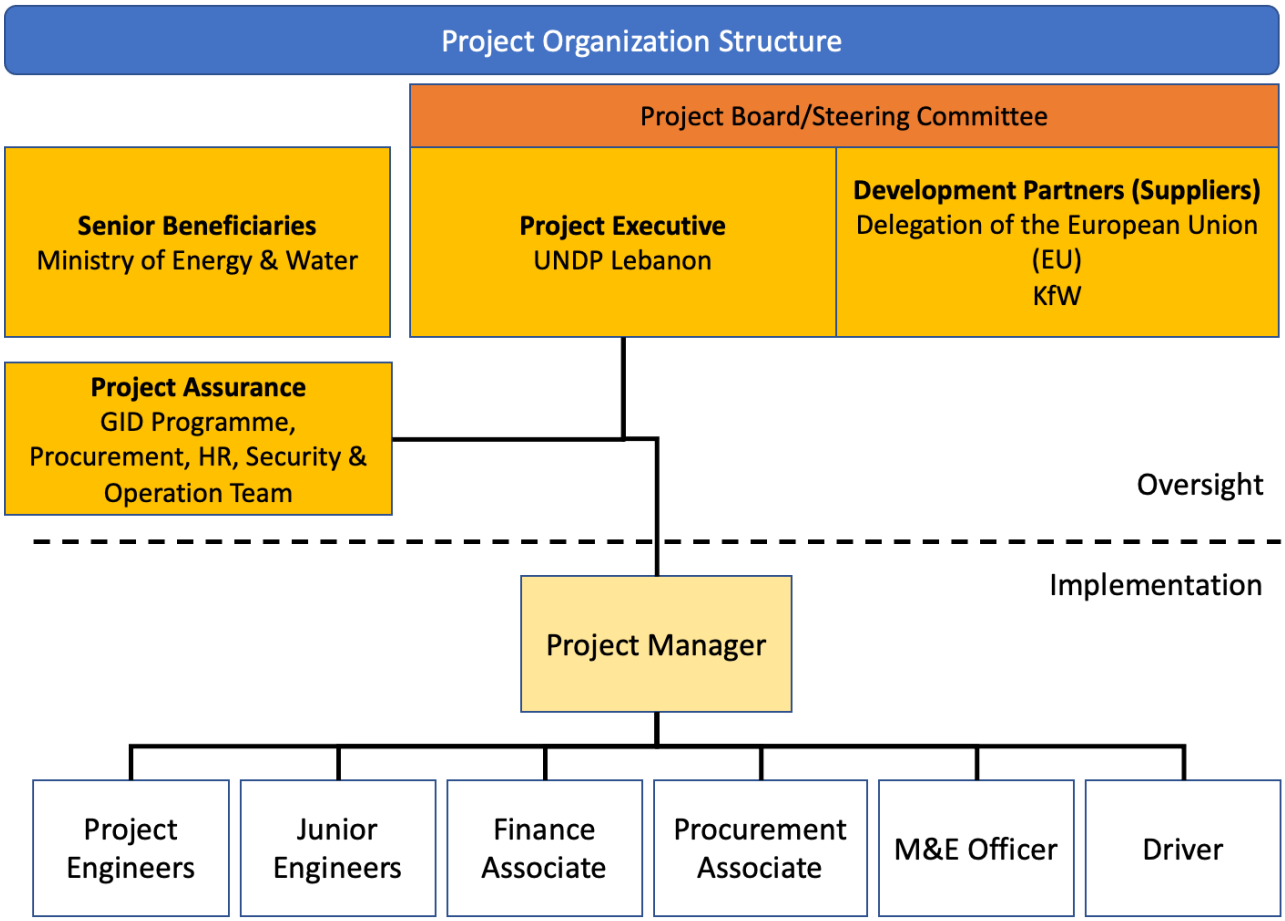
Given that this Action will be implemented in parallel to and in full coordination with another project funded by the Government of Germany through the German Development Bank, the Action will be governed by the same structure and reporting will be synchronised and aligned to the extent possible while keeping separate reporting documentation because each project has a separate agreement. Activities will also be coordinated between the projects while maintaining separate and distinct budgets. Some of the project team members will be shared between the two projects for purposes of efficiency, including the Project Manager position.

A 'Project Board' will be set up and will be responsible for making, by consensus, management decisions for the project when guidance is required by the Project Manager, including recommendation for UNDP approval of project plans and revisions. The Project Board will meet every year or more frequently as needed by the project. The Project Board will also provide expertise and ensure that the various interventions are in line with the national priorities and are well coordinated with other on-going activities within the sector.

In specific, the responsibilities of the Project Board include:

- Provide overall guidance and direction to the project, ensuring it remains within any specified constraints
- Address project issues as raised by the project manager
- Provide guidance on new project risks, and agree on possible countermeasures and management actions to address specific risks
- Agree on project manager's tolerances as required
- Review the project progress, and provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to plans
- Appraise the annual project implementation report, including the quality assessment rating report; make recommendations for the work-plan
- Provide ad hoc direction and advice for exceptional situations when the project manager's tolerances are exceeded
- Assess and decide to proceed on project changes through appropriate revisions

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Project Management and Office Costs

The titles and roles of all personnel are described in Table 2 below. The related cost are included within output 1.1. in the project budget.

Table 2. Project and Country Office Personnel Overview

Position	No.	Status	Key responsibilities
Project Manager	1	Part-time (50%)	The Project Manager will be responsible for day-to-day management, financial, administrative & technical control. Will coordinate between the various results and will be the focal point to key stakeholders. Will also provide advisory support on the topic of the Action
Project Coordinator	1	Full-time	The Project Coordinator will help in the coordination of the project engineers and the site engineers to support all the implementation of activities and liaise with the Project Manager, M&E Associate and Procurement Associate for the overall procurement and contract management for the various implementation activities.
Project Engineer	4	Full-time	The Project Engineers will be responsible for the oversight of the design of the installation, bidding processes, specifications setting, follow-up on site with beneficiaries, data gathering and reporting for each of the sectors where works are being implemented
Site Engineer	3	Full-time	The site engineers will support the project engineers in site supervision of contractors and follow-up on the installation

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			of all installations
Procurement Associate	1	Full time	The Procurement Associate will support in the implementation of project procurement processes, advising in procurement evaluation processes, supporting in negotiations with potential contractors (as needed), assisting in the process of contracting, monitoring of contracts.
Project Associate	1	Full-time	The project Project Associate will provide administrative and logistical support & will liaise with the Country Office on financial, administrative and operational activities.
M&E Associate	1	Full time	The project Monitoring & Evaluation Associate will work on consolidating, analysing and gathering all project results and data to better report and provide analysis needed for communication, policy and technical support to the project and the Country Office.
Driver	1	Full-time	The Drivers will be responsible for the missions undertaken within the project. Will maintains vehicle logs & the vehicles.
Programme Manager	1	Part-time (20%)	The Programme Manager will have the responsibility of project quality assurance and technical project oversight. Will assist the project manager in progress reports, overall management of project risks and oversight of the implementation of the project activities.
Programme Associate	1	Part-time (20%)	The Programme Associate will be responsible for oversight of project operations, financial duties, financial monitoring and reporting and assist the project in financial transactions and related project reports, processing of payments and management of project operations such as office logistics, management of equipment, etc.
Procurement Manager	1	Part-time (15%)	Head of procurement will be responsible for the quality assurance on all procurement cases and will undertake contract management as needed, particularly in terms of negotiations and adhoc legal challenges or problems
Operations Manager	1	Part-time (15%)	Operations Manager will be responsible for administrative quality assurance, advising and verifying procurement and human resources processes, and managing external relations related to all operational aspects of project.
Security Officer	1	Part-time (10%)	The Security Officer will ensure safety and security of the project staff during site visits and missions to the location of the project activities, provide advice regarding the overall security and political situation governing the country
Programme Specialist	1	Part-time (10%)	Programme Specialist staff to oversight the reporting, financial management, legal aspects and the partnership with the EU from the UNDP Brussels Office.

GMS/F&A

A rate of 7% will be applied to EU funds; A reate of 8% will be applied to the UNDP 888 funds.

General Expenses

The Budget for the Action also provides for the office rental costs, assets for the operation of the office, maintenance and repair contracts, furniture, consumables and supplies for the operation of the office, costs of IT and telecommunication services, costs of electricity and water, facility management, vehicles costs and insurance costs as well as local and international travel. The related cost are included within output 1.1. in the project budget.

IX. LEGAL CONTEXT

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Lebanon and UNDP, signed in 1986. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by UNDP in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

X. RISK MANAGEMENT

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the [project funds][20] are used to provide support to individuals or entities associated with terrorism , that the recipients of any amounts provided by UNDP hereunder do not appear on the United Nations Security Council Consolidated Sanctions List, and that no UNDP funds received pursuant to the Project Document are used for money laundering activities. The United Nations Security Council Consolidated Sanctions List can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. In the implementation of the activities under this Project Document, UNDP as the Implementing Partner will handle any sexual exploitation and abuse ("SEA") and sexual harassment ("SH") allegations in accordance with its regulations, rules, policies and procedures.

6. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

7. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor, and sub-recipient:

a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:
   i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

[20] To be used where UNDP is the Implementing Partner

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. Each responsible party, subcontractor and sub-recipient (each a "sub-party" and together "sub-parties") acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the sub-parties, and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.
(a) In the implementation of the activities under this Project Document, each sub-party shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").
(b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, each sub-party, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment. SH may occur in the workplace or in connection with work. While typically involving a pattern of conduct, SH may take the form of a single incident. In assessing the reasonableness of expectations or perceptions, the perspective of the person who is the target of the conduct shall be considered.

d. In the performance of the activities under this Project Document, each sub-party shall (with respect to its own activities), and shall require from its sub-parties (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, sub-parties will and will require that their respective sub-parties will take all appropriate measures to:
(i) Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;
(ii) Offer employees and associated personnel training on prevention and response to SH and SEA, where sub-parties have not put in place its own training regarding the prevention of SH and SEA, sub-parties may use the training material available at UNDP;
(iii) Report and monitor allegations of SH and SEA of which any of the sub-parties have been informed or have otherwise become aware, and status thereof;
(iv) Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and
(v) Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. Each sub-party shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the relevant sub-party shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

e. Each sub-party shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the relevant sub-party to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

f. Each responsible party, subcontractor and sub-recipient will ensure that any project activities undertaken by them will be implemented in a manner consistent with the UNDP Social and Environmental Standards and shall ensure that any incidents or issues of non-compliance shall be reported to UNDP in accordance with UNDP Social and Environmental Standards.

g. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud, corruption or other financial irregularities, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption, anti-fraud and anti money laundering and countering the financing of terrorism policies are in place and enforced for all funding received from or through UNDP.

h. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a) UNDP Policy on Fraud and other Corrupt Practices (b) UNDP Anti-Money Laundering and Countering the Financing of Terrorism Policy; and (c) UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

i. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

j. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud, corruption other financial irregularities with due confidentiality.
Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

k. UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud corruption or other financial irregularities, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement.

Where such funds have not been refunded to UNDP, the responsible party, subcontractor or sub-recipient agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud, corruption or other financial irregularities, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

l. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

m. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

n. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled "Risk Management Standard Clauses" are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.

XI. ANNEXES

1. **Project Quality Assurance Report (done separately in system)**
2. **Social and Environmental Screening Template**
3. **Risk Analysis**. Use the standard Risk Register template. Please refer to the Deliverable Description of the Risk Register for instructions
4. **Capacity Assessment:** Results of capacity assessments of Implementing Partner (including Partner Capacity Assessment Tool (PCAT) and HACT Micro Assessment) – PCAT and Micro-Assessment completed separately.
5. **On-Granting Provisions Applicable to the Implementing Partner[21].** Not Applicable

[21] Applicable for non-UNDP Implementing Partner as Grant Making Institution facilitating on-granting.

Annex 2 Social and Environmental Screening Template

UNDP SOCIAL AND ENVIRONMENTAL SCREENING TEMPLATE (V. JULY 2022)

The completed template, which constitutes the Social and Environmental Screening Report, must be included as an annex to the Project Document at the design stage. Note: this template will be converted into an online tool. The online version will guide users through the process and will embed relevant guidance.

Project Information

Project Information	
1. Project Title	Sustainable Energy for Public Services entities, entrepreneurship and innovation– EU Support to Lebanon's Green Agenda (CEDRO 6)
2. Project Number (i.e. Atlas project ID, PIMS+)	01001256
3. Location (Global/Region/Country)	Lebanon
4. Project stage (Design or Implementation)	Design
5. Date	Sep 2023

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
The project consists of approaching the transition to renewable energy in Lebanon at two levels: one is to provide and improve access to energy for the public sector, including the Lebanese Armed Forces and the Internal Security Forces and the second is to support the private sector to upscale solar energy supply to greening recovery and creating jobs. Parallel to these two approaches, improving capacities, policies and legislation to push the sector forward will also be tackled based on the lessons learnt and needs identified during implementation at the field level of both components. Ultimately, the overall impact of the project is the improved energy security and transition to sustainable energy and resource efficiency in Lebanon. Thus, the project seeks to further the realization of human rights by improving the access and availability of energy for all populations in Lebanon, which is fundamental to any human rights. There are no potential adverse impacts of this project that might affect human rights as the project will result in the further promotion of renewable energy solutions and the reduction of environmental degradation and its negative impacts on public health.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
The challenge of achieving sustainable development is intricately linked with the need for gender equality. Unfortunately, a gender gap exists in the renewable energy sector in Lebanon, particularly in terms of work opportunities for women in the energy value chain. Building on UNDP's previous work to bridge the gap in the field of gender and energy as well as climate change action, gender aspects will be mainstreamed across the various activities in this Action. This will include the promotion of women in the solar energy vocational programmes and the encouragement of their active participation in training to encourage women technicians and installers in the sector. Gender considerations will

also be integrated into the innovation activities to encourage young women entrepreneurs to participate in sustainable energy and climate start-ups and ideas. Women start-ups will be encouraged as well as women-led private sector beneficiaries within this Action. Furthermore, the project will engage with policymakers, advocating for gender-inclusive energy policies and frameworks. By highlighting the social, economic, and environmental benefits of gender-responsive energy solutions, it aims to create an enabling environment for change. Ultimately, the project strives to create a future where the involvement of youth and women in the sustainable energy and climate challenge is fully integrated. By championing gender equality in the renewable energy sector, it aims to unlock the potential of women, promote economic empowerment, and pave the way for a more sustainable and equitable world.
Briefly describe in the space below how the project mainstreams sustainability and resilience
Environmental benefits of the renewable energy systems to be installed target mostly the reduction of pollution from self-generation diesel use, mainly carbon dioxide (a green-house gas – GHG), carbon monoxide, particulate matter, nitrogen oxides, and sulphur dioxide among other. They also will target delivering cleaner energy in terms of kWh than that of the Lebanese grid, which is mostly composed of fuel oil-generated power. In some occasions, the technologies selected will enable the complete shutting down of diesel generators, reducing thus the level of noise pollution as well. Therefore, the project not only targets climate mitigation action but also has environmental sustainability benefits and promotes the integration of SDG goals such as gender and sustainable cities as well.
Briefly describe in the space below how the project strengthens accountability to stakeholders
The project design will involve close interaction and participation of stakeholders mainly, government, municipalities, and the general public etc. The project will raise awareness among project stakeholders and create channels for communication through website, social media, telephone and government and project offices. The project will also disseminate information through online and printed media for relevant stakeholders to access in all languages to ensure that everyone is updated on project developments. The project website will also accommodate grievances and a redress mechanism will also be put in place. In addition, UNDP's Accountability Mechanism, which includes the Social and Environmental Compliance Review (SECU) and Stakeholder Response Mechanism (SRM) will also serve as an additional layer of grievance redress and empower stakeholders to push for accountability.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
Risk1: Marginalization of vulnerable groups when selecting the sites for environmental projects and installation of solar PV systems Related to: • Human Rights; P.5, P.6 • Accountability; P.13, P.14	I = 2 L = 3	Low	Site selection for installation of solar PV may marginalize stakeholders from participating in this sector, or from having equal level of access to energy if selection is not done in collaboration with all stakeholders. On the other hand, the project will focus on public institution-level support (e.g. public university, hospitals) based on the vulnerability assessment to benefit the targeted community as broadly as possible. In such intervention, marginalization of specific group is much less likely although the clear communication is still needed. Therefore, the significance of this risk is rated at 'low'.	The target groups of the project are mainly the more vulnerable people that access to the public institutional services and the LAF. The project GRM will also help in managing this risk by providing a means for affected stakeholders to raise concerns and/or grievances.
Risk 2: Reproducing existing discriminations against women through excluding them from decision-making on project activities, benefiting from project outputs and capacity building initiatives Related to: • Gender Equality and Women	I = 3 L = 3	Moderate	The male dominated nature of the energy sector and the limited social status and opportunities identified for women. UNDP's study identified lack of STEM education awareness for women as one of the challenges in Lebanon (2016). According to the study, women participation in	Gender-balanced representation in the national dialog, deliberate inclusion of electricity end-uses prioritized by community women at the feasibility stage, and outreach and training support to women, youth, and the financial institutions that serve them are examples of actions meant to prevent unbalanced appropriation by men of solar PV technology, services, and benefits.

Empowerment; P.10			various social, economic and political activities remained low with a female participation of "23.5% compared to 70.3% for men". This may pose a challenge to ensure that women will have the chance to participate at the decisions-making level.	
Risk 3: Climate events and disasters (including floods) on the project sites Related to: • Standard 2: Climate Change and Disaster Risks; 2.2 • Standard 3: Community Health, Safety and Security; 3.3	I = 2 L = 2	Low	Extreme weather events due to climate change might affect infrastructure in Lebanon. Intense precipitation amounts can lead to the flooding of infrastructure. All solar PV systems are by-nature open air structures exposed to climate events and involve built structures that may be vulnerable to the impacts of such an event. Due to the fact that the sites have not yet been selected, the likelihood of this risk is not known, and a conservative approach is adopted.	UNDP undertook the flood risk management and mapping fpr Lebanon (2016). Based on this assessment, the project will assess the flood risk for the potential sites to minimize the risk exposure.
Risk 4: Working conditions not in line with national and international standards (by contractor or other entities involved in the project) Related to: • Standard 7: Labor and Working Conditions; 7.1, 7.2, 7.3, 7.4, 7.5, 7.6	I = 3 L = 2	Moderate	Installation of solar PV or other environmental approaches will require labour, some of which may be sourced to unskilled/manual labourers who could be less familiar with the type of installations considered for this project and the concomitant occupational health and safety requirements and risks. In addition, manufacturers, suppliers, subcontractors and subcontractors within the solar supply chain may not be in line with SES and thus procurement of solar PV may contribute to	For each procurement, labour standards, social safeguards and an Occupational Safety and Health (OSH) measures will be applied for the project to ensure labour standards and rights are upheld for project workers. The Contractors will be closely monitored by the UNDP during the project implementation for their adherence to these standards.

			working conditions that undermine worker human rights, health and safety.	
Risk 5 Generation of hazardous waste (specifically e-waste) from the installed solar PV Related to: • Standard 8: Pollution Prevention and Resource Efficiency; 8.2	I = 2 L = 4	Moderate	Very modest amounts of waste will be generated during installation of solar PV (ground movement and concrete residues); electric wiring and insulator ends; broken or rejected parts and components. At the end of its lifespan, electronic waste ("e-waste") in the form of solar panels and/or batteries will be generated. Without proper handling directives, disposal and/or recycling mandate for obsolete equipment, this could result in additional waste generation, including of hazardous/phase-outs materials, chemicals or other pollutants (e.g. from batteries). Failure to recycle non-hazardous waste could also contribute to additional waste generation.	Lebanon has no functioning formal waste management infrastructure for e-waste and batteries. To address this issue, the UNDP is now implementing the EU-funded project 'Decentralised Waste Management Integrated Response (2021-2024)'. The project aims at reaching all of the sorted material to be collected and a maximum recycling of E-waste and batteries through a system of EPR, preferably to be implemented voluntarily by the sector within constraints set by the national government and in close cooperation with the municipalities. Therefore, the above UNDP's project will serve as mitigation measures for this project.

	QUESTION 4: What is the overall project risk categorization?		
	Low Risk	☐	
	Moderate Risk	**X**	5 potential risks have been identified for this project, three of which are assessed as MODERATE and three as LOW. As a result, this project is rated overall as a MODERATE Risk project. During the project design, management measures has been incorporated into ProDoc.
	Substantial Risk	☐	
	High Risk	☐	

	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)				
	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")				***Status? (completed, planned)***
	if yes, indicate overall type and status			Targeted assessment(s)	
				ESIA (Environmental and Social Impact Assessment)	
				SESA (Strategic Environmental and Social Assessment)	
	Are management plans required? (check if "yes)				
	If yes, indicate overall type			Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	

				ESMP (Environmental and Social Management Plan which may include range of targeted plans)	
				ESMF (Environmental and Social Management Framework)	
	Based on identified risks, which Principles/Project-level Standards triggered?		**Comments (not required)**		
	Overarching Principle: Leave No One Behind				
	Human Rights	X			
	Gender Equality and Women's Empowerment	X			
	Accountability	X			
	1. Biodiversity Conservation and Sustainable Natural Resource Management				
	2. Climate Change and Disaster Risks	X			
	3. Community Health, Safety and Security	X			
	4. Cultural Heritage				
	5. Displacement and Resettlement				
	6. Indigenous Peoples				
	7. Labour and Working Conditions	X			
	8. Pollution Prevention and Resource Efficiency	X			

Final Sign Off

Final Screening at the design-stage is not complete until the following signatures are included

Signature	*Date*	*Description*
QA Assessor Jihan Seoud Programme Manager DocuSigned by: B9C552CD991843C...	22-Sep-2023	UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver Mohammed Salih Deputy Resident Representative DocuSigned by: 4C17B57AEAA5467...	22-Sep-2023	UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair Mohammed Salih Deputy Resident Representative DocuSigned by: 4C17B57AEAA5467...	22-Sep-2023	UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

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SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.		
Overarching Principle: Leave No One Behind **Human Rights**		**Answer (Yes/No)**
P.1	Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
P.2	Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	Yes
P.3	Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:		
P.4	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
P.5	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [16]	Yes
P.6	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	Yes
P.7	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment		
P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	No
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		

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P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes
P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	No
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	No
1.7	adverse impacts on soils?	No
1.8	harvesting of natural forests, plantation development, or reforestation?	No
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.12	handling or utilization of genetically modified organisms/living modified organisms?[17]	No
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[18]	No
1.14	adverse transboundary or global environmental concerns?	No
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	No
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	No
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	No

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3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	No
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	Yes
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	No
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	No
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	No
3.7	influx of project workers to project areas?	No
3.8	engagement of security personnel to protect facilities and property or to support project activities?	No
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	No
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.4	alterations to landscapes and natural features with cultural significance?	No
4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	risk of forced evictions?[19]	No
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	No
6.2	activities located on lands and territories claimed by indigenous peoples?	No
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then Standard 6 requirements apply, and the potential significance of risks related to impacts on indigenous peoples must be Moderate or above.* [*]	No

[*] Note: revised July 2022 modifying presumption of risk significance from Substantial or higher to Moderate or higher.

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6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	No
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	No
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labour laws and international commitments?	Yes
7.2	working conditions that may deny freedom of association and collective bargaining?	Yes
7.3	use of child labour?	Yes
7.4	use of forced labour?	Yes
7.5	discriminatory working conditions and/or lack of equal opportunity?	Yes
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	No
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	No
8.5	the application of pesticides that may have a negative effect on the environment or human health?	No
8.6	significant consumption of raw materials, energy, and/or water?	No

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Annex 3 Risk Analysis

OFFLINE PROJECT RISK REGISTER TEMPLATE

Project Title:	**Project Number: 01001256**	**Date: prodoc signature**

#	Event	Cause	Impact(s)	Risk Category and Sub-category *(including Risk Appetite)*	Impact, Likelihood & Risk Level *(see Annex 3 Risk Matrix)*	Risk Valid From/To	Risk Owner *(individual accountable for managing the risk)*	Risk Treatment and Treatment Owner
1	There is a risk of damage to installed equipment and material …	As a result of beneficiaries not maintaining or protecting the system	Which will impact in lack of sustainability of equipment/real benefit	**3. OPERATIONAL (3.5. Partners' engagement) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 1.1: Memorandums of understanding will be signed with beneficiaries to ensure the protection, operation and maintenance of the equipment installed…. Risk Treatment Owner: Project Manager
2	There is a risk that the solar systems will be damaged by natural disasters or environmental degredation	As a result of natural disasters or environmental degredation	Which will impact on their sustainability/operation ….	**1. SOCIAL AND ENVIRONMENTAL (1.5. Climate change and disaster risks) - UNDP Risk**	Likelihood: **1 - Not likely** Impact: **3 - Intermediate**	From: 01-Nov-23 To: 31-Oct-27	… UNDP	Risk Treatment 2.1: Ensure all outdoor equipment can withstand severe weather conditions expected in Lebanon Risk Treatment Owner: Project Manager

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				Appetite: CAUTIOUS	Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**			Risk Treatment 2.2: Ensure that an environmental statement is undertaken for all projects with significant installations.… Risk Treatment Owner: Project Manager
3	There is a risk that at the end of the project the batteries may need to be disposed	As a result of duration/lifetime of the technology	Which will impact in solid waste	**1. SOCIAL AND ENVIRONMENTAL (1.4. Biodiversity conservation and sustainable natural resource management) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite ofEXPLORATORY)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 3.1: UNDP will be working on battery disposal through the EU-funded TaDWIR project. Activities will be coordinated with this Action. … Risk Treatment Owner: Project Manager

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4	There is a risk that the beneficiaires will not have the Ability to effectively operate and maintain installed systems …	As a result of lack of financing and/or capacity…	Which will impact in the operation of the installed systems….	**3. OPERATIONAL (3.5. Partners' engagement) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **4 - Highly likely** Impact: **4 - Extensive** Risk level: **SUBSTANTIAL (equates to a risk appetite of OPEN)**	From: 01-Nov-23 To: 31-Oct-27	… UNDP	Risk Treatment 4.1: A minimum of two-years performance guarantee for all systems installed will be required. The possibility of including O&M contract will also be assessed and capacity building activities will be ensured.… Risk Treatment Owner: Project Manager

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5	There is a risk that Delays in import clearance will cause delays to the project …	As a result of institutional capacity of government entities	Which will impact in delay in project implementation	**3. OPERATIONAL (3.5. Partners' engagement) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Nov-23 To: 31-Oct-27	… *UNDP*	Risk Treatment 5.1: Sufficient time will be given for project implementation to account for expected delays in manufacturing, shipping and clearance of imported system components … Risk Treatment Owner: Project Manager

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6	There is a risk of Delays in manufacturing and shipping of equipment …	As a result of external factors …	Which will impact in…delay in project implementatio	**3. OPERATIONAL (3.8. Capacities of the partners) - UNDP Risk Appetite: EXPLORATORY TO OPEN**	Likelihood: **2 - Low likelihood** Impact: **3 - Intermediate** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Nov-23 To: 31-Oct-27	…	Risk Treatment 6.1: Continuous coordination will be conducted with the contractors and suppliers to overcome any possible delays in manufacturing or shipping…. Risk Treatment Owner: Project Manager

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7	There is a risk that there is Instability in the security situation in the country …	As a result of the security situation in Lebanon	Which will impact in delays in project implementation	**8. SAFETY AND SECURITY (8.1. Armed conflict) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 7.1: Isolated security incidences may occur. Appropriate measures will be taken accordingly. These could include the postponement of some of the works on site, and the coordination with the Internal Security Forces and UN Department of Security as needed… Risk Treatment Owner: Project Manager

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8	There is a risk that Availability of funds to assist in co-financing private sector contribution	As a result of the financial situation in Lebanon	Which will impact in delays in project implementation	**2. FINANCIAL (2.4. Fluctuation in credit rate, market, currency) - UNDP Risk Appetite: MINIMAL TO CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **4 - Extensive** Risk level: **SUBSTANTIAL (equates to a risk appetite of OPEN)**	From: 01-Nov-23 To: 31-Oct-27	UNDP…	Risk Treatment 8.1: Available funds will be sought, such as CEDAR Oxygen, and support from other institutions (e.g. Cross Boundary … Risk Treatment Owner: Project Manager
9	There is a risk that Capital controls and the inability to raise bank guarantees …	As a result of the financial situation in Lebanon	Which will impact in achieving project targets	**2. FINANCIAL (2.4. Fluctuation in credit rate, market, currency) - UNDP Risk Appetite: MINIMAL TO CAUTIOUS**	Likelihood: **3 - Moderately likely** Impact: **3 - Intermediate** Risk level: **MODERATE (equates to a risk appetite of EXPLORATORY)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 9.1: Contractual agreements to be signed between the Implementing partners and the private sector beneficiaries that would overcome banking challenges to the extent possible…. Risk Treatment Owner: Project Manager

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10	There is a risk that marginalization of vulnerable groups when selecting the sites for environmental projects and installation of solar PV systems	As a result of the site selection of the project	Which will impact in negative perception or poor targeting	**5. REPUTATIONAL (5.5. Stakeholder management) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **2- Low** Impact: **3 - Intermediate** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 10.1: The target groups of the project are mainly the more vulnerable people that access the public institutions and the LAF Risk Treatment Owner: Project Manager
11	There is a risk **Reproducing existing discriminations against women through excluding them from decision-making on project activities, benefiting from project outputs and capacity building initiatives** …	As a result of the approach of project implementation	Which will impact in negative perception or poor targeting	**1. SOCIAL AND ENVIRONMENTAL (1.2. Gender equality and women's empowerment) - UNDP Risk Appetite: CAUTIOUS**	Likelihood: **2- Low** Impact: **3 - Intermediate** Risk level: **LOW (equates to a risk appetite of CAUTIOUS)**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 10.1: Gender-balanced representation in the national dialog, deliberate inclusion of electricity end-uses prioritized by community women at the feasibility stage, and outreach and training support to women, youth, and the financial institutions that serve them are examples of actions meant to prevent unbalanced appropriation by men of solar PV technology, services, and benefits. Risk Treatment Owner: Project Manager
12	There is a risk that **Working conditions not in line with national and international standards (by contractor or other entities involved in the project)**	As a result of the contracting works	Which will impact in negative labour impacts	**1. SOCIAL AND ENVIRONMENTAL (1.10. Labour and working conditions) - UNDP** Risk Appetite: CAUTIOUS	Likelihood: **2- Low** Impact: **3 - Intermediate** Risk level: **LOW (equates to a**	From: 01-Nov-23 To: 31-Oct-27	UNDP	Risk Treatment 11.1 For each procurement, labour standards, social safeguards and an Occupational Safety and Health (OSH) measures will be applied for the project to ensure labour standards and rights are upheld for project workers. Risk Treatment Owner: Risk Treatment 11.2 The Contractors

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					risk appetite of CAUTIOUS)			will be closely monitored by the UNDP during the project implementation for their adherence to these standards. Risk Treatment Owner: Project Manager